

Indian Statistical Institute

BSDS IInd Year

Academic Year 2025 - 2026: Semester I

Course: Probability II

Instructor: Antar Bandyopadhyay

Assignment # 9

Date Given: October 22, 2025

Date Due: October 31, 2025
Total Points: 10

1. Suppose $1 \leq p \leq p'$ then show that $\mathcal{L}_{p'} \subseteq \mathcal{L}_p$.
2. Show by example that for any pair $1 \leq p \leq p'$ one can always have a random variable, say X , such that, $X \in \mathcal{L}_p$ but, $X \notin \mathcal{L}_{p'}$.
3. For $Z \sim \text{Normal}(0, 1)$ find $\|Z\|_p$ in terms of $p \in \mathbb{N}$.
4. Is it possible to have a random variable, say X , such that, $\mathbf{E}[|X|^p] = \infty$ for any $p > 0$?